SUMMARY OF SCATTERING

- Scattering wave function: $\varphi^{SCATT}(\vec{r})\Big|_{r\to\infty} = e^{ikz} + f_k(\theta,\varphi)\dfrac{e^{ikr}}{r}$.

  Scattering amplitude.

- Schrödinger: $\left[\nabla^2 + k^2 - u(r)\right]\varphi(r) = 0$

  with $u(r) = \dfrac{2\mu}{\hbar^2}V(r)$,

  $E = \dfrac{\hbar^2 k^2}{2\mu}$.

- Probability Current: $j(\vec{r},t) = i\dfrac{\hbar}{2\mu}\left[\psi\nabla\psi^* - \psi^*\nabla\psi\right]$, $\rho = |\psi|^2$

  obeys continuity $\dfrac{\partial\rho}{\partial t} + \nabla\cdot\vec{j} = 0$.

- Cross-section: $d\Omega = \sin\theta\, d\theta\, d\varphi$ "solid angle"

$$d\sigma(\theta,\varphi) \sim \frac{\#\text{ scattered particles through } d\Omega}{\text{incoming flux}}$$

$$= |f_k(\theta,\varphi)|^2\, d\Omega$$

$$\frac{d\sigma(\theta,\varphi)}{d\Omega} = |f_k(\theta,\varphi)|^2 \quad \text{"differential cross-section"}.$$

$$= \frac{\#\text{ events in } d\Omega}{\text{unit time}}$$

- Born Approximation: $(\nabla^2 + k^2)\varphi(r) = u(r)\,\varphi(r)$

  integral eqn $\longrightarrow$ $(\nabla^2 + k^2)\,G_1(\vec{r}-\vec{r}\,') = \delta^{(3)}(\vec{r})$.

E & M:

$$-\nabla^2\left(\frac{1}{4\pi r}\right) = \delta(r)$$

Solved by $G_\pm = -\frac{1}{4\pi r} e^{\pm ikr}$

Here $\psi^{SCATT}(\vec{r})\Big|_{r\to\infty} = \psi_0(\vec{r}) + \int d^3\vec{r}\,' \, G_+(\vec{r}-\vec{r}\,') \, \psi(\vec{r}\,') \, u(\vec{r}\,')$

+ higher contributions ...
keep iterating

$$= e^{ikz} + \int d^3\vec{r}\,' \, G_+(\vec{r}-\vec{r}\,') \, u(\vec{r}\,') \, e^{i\vec{k}\cdot\vec{r}\,'}$$

[LEADING ORDER]

$$= e^{ikz} + \frac{e^{ikr}}{4\pi r} \int d^3r \, u(\vec{r}) \, e^{i\vec{r}\cdot(\vec{k}^{(in)} - \vec{k}^{(out)}(\theta,\varphi))}$$

and $f_k(\theta,\varphi) = -\frac{1}{4\pi} \int d^3r \, u(\vec{r}) \, e^{i\vec{r}\cdot(\vec{k}^{(in)} - \vec{k}^{(out)}(\theta,\varphi))}$.

Arises from $G_+(\vec{r}-\vec{r}\,')$.

Can invoke $\vec{k}' = \vec{k}^{(in)} - \vec{k}^{(out)}$ if V θ/φ-independent. Then do $\vec{k}'\cdot\vec{r}$.

$$k' = k(1-\cos\theta) \rightarrow 2k\sin^2\left(\frac{\theta}{2}\right)$$

→ Is a good approximation when "Small term" << "leading term"

in the range of the potential V:

$$\left|e^{ikz}\right| \gg \left|\frac{e^{ikr}}{r} f_k(\theta,\varphi)\right|$$

- Yukawa Potential: $V(r) = \frac{A}{r} e^{-\mu r}$.

- Review of Spherical Harmonics, Legendre Polynomial Identities

$$\rightarrow \int_{-1}^{1} P_\ell(x) P_{\ell'}(x)\, dx = \int_0^{\pi} P_\ell(\cos\theta) P_{\ell'}(\cos\theta) \sin\theta\, d\theta = \frac{2\delta_{\ell\ell'}}{2\ell+1}.$$

$$\rightarrow \int_0^{2\pi}\int_0^{\pi} \sin\theta\, d\theta\, d\varphi\; Y_\ell^{m*}(\theta,\varphi)\, Y_{\ell'}^{m'}(\theta,\varphi) = \delta_{mm'}\delta_{\ell\ell'}$$

$$P_\ell(x) = \frac{1}{2^\ell \ell!}\frac{d^\ell}{dx^\ell}[x^2-1]^\ell$$

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x)$$ Associated Legendre Polynomials

$$Y_\ell^m(x) = N e^{im\varphi} P_\ell^m(\cos\theta)$$ Spherical Harmonics

$$Y_\ell^0 = \sqrt{\frac{2\ell+1}{4\pi}}\, P_\ell(\cos\theta)$$ [$m=0$; potential is φ-independent]

$$j_\ell(x) = (-1)^\ell x^\ell \left(\frac{1}{x}\frac{d}{dx}\right)^\ell \left[\frac{\sin x}{x}\right]$$ Spherical Bessel of 1^{st} kind

$n_\ell(x)$ = "Neumann Function"

$$h_\ell^{(1)}(x) = j_\ell(x) + i\, n_\ell(x) \quad ; \quad h_\ell^{(2)} = h_\ell^{(1)*}.$$

Any function can be expanded into a basis of P_ℓ's, Y_ℓ^m's, and $j_\ell / n_\ell / h_\ell^{(i)}$'s.

$$e^{i\vec{k}\cdot\vec{r}} = 4\pi \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} i^\ell\, Y_\ell^{m*}(\theta,\varphi)\, j_\ell(kr)\, Y_\ell^m(\theta,\varphi)$$

More importantly

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^\ell (2\ell+1)\, j_\ell(kr) P_\ell(\cos\theta)$$

- Angular momentum: $\hat{L}^2 \varphi = \hbar^2 \ell(\ell+1) \varphi$

$$\hat{L}_z \varphi = \hbar m \varphi \qquad \varphi \equiv \text{eigenstate}$$

- Partial Waves:

$$j_\ell(kr) \overset{r\to\infty}{\sim} \frac{\sin(kr - \frac{\ell\pi}{2})}{kr}$$

$$n_\ell(kr) \overset{r\to\infty}{\sim} \frac{\cos(kr - \frac{\ell\pi}{2})}{kr}$$

The solution to the Bessel Differential Equation

$$r^2 \frac{d^2 f_{k\ell}}{dr^2} + 2r \frac{df_{k\ell}}{dr} + (k^2r^2 - \ell(\ell+1)) f_{k\ell} = k^2 f_{k\ell}$$

is $\quad f_{k\ell}(r) = \alpha j_\ell(kr) + \beta n_\ell(kr)$

Review of Derivations:

Probability current: Let $\rho = \psi^* \psi$, where ψ obeys the schrödinger equation

$$i\hbar \frac{\partial}{\partial t} \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(r)\psi,$$

and let $V = 0$.

$$\frac{d\rho}{dt} = \frac{\partial \psi^*}{\partial t} \psi + \psi^* \frac{\partial \psi}{\partial t}$$

$$= \left(\frac{-i\hbar}{2m} \nabla^2 \psi^* \right) \psi + \psi^* \left(\frac{i\hbar}{2m} \nabla^2 \psi \right)$$

$$= \frac{i\hbar}{2m} \left[\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right].$$

But observe that $\psi^* \partial_i^2 \psi = \partial_i (\psi^* \partial_i \psi) - \partial_i \psi^* \partial_i \psi$

hence

$$= \frac{i\hbar}{2m} \left[\nabla(\psi^* \nabla \psi) - \nabla \psi^* \nabla \psi - \nabla(\psi \nabla \psi^*) + \nabla \psi \nabla \psi^* \right]$$

$$= \frac{i\hbar}{2m} \nabla \cdot \left[\psi^* \nabla \psi - \psi \nabla \psi^* \right]$$

$$= \frac{\partial \rho}{\partial t}$$

Hence $\dfrac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$

where $\vec{j} = \dfrac{i\hbar}{2m} \left[\psi \nabla \psi^* - \psi^* \nabla \psi \right]$ denotes the probability current.

Alternate definition:

Write $\psi = |\psi| e^{i \arg \psi}$. Then

$$\vec{j} = \frac{i\hbar}{2m}\Big[|\psi| e^{i\arg\psi}\, \underbrace{\cancel{\nabla(|\psi|)}}_{0}\, e^{-i\arg\psi} + |\psi| \cancel{e^{i\arg\psi}}\, |\psi|^* (-i\nabla\arg\psi)\, \cancel{e^{-i\arg\psi}}$$

$$- |\psi|^* e^{-i\arg\psi}\, \underbrace{\cancel{\nabla(|\psi|)}}_{0}\, e^{i\arg\psi} - |\psi|^* \cancel{e^{-i\arg\psi}}\, |\psi| \,(i\nabla\arg\psi)\, \cancel{e^{i\arg\psi}}\Big]$$

$$= \frac{i\hbar}{2m}\Big[|\psi|^2(-i\nabla\arg\psi) - |\psi|^2(i\nabla\arg\psi)\Big]$$

$$= \frac{\hbar}{m}|\psi|^2\, \nabla\arg\psi .$$

Green's Function Operator:

Solve the radial SE $\left(\nabla^2 + k^2 - u(\vec{r})\right) G_\pm(\vec{r}-\vec{r}\,') = \delta^{(3)}(\vec{r}-\vec{r}\,')$

The solution is $\varphi(\vec{r}) = \varphi_0 + \int d^3\vec{r}\,'\, G_\pm(\vec{r}-\vec{r}\,')\, u(\vec{r}\,')\, \varphi(\vec{r}\,')$

which is iterated over to obtain the Born series.

The solution to $G_\pm(\vec{r}) = -\frac{1}{4\pi}\frac{e^{\pm i\vec{k}\cdot\vec{r}}}{|r|}$, since $\nabla^2\left(-\frac{1}{4\pi r}\right) = \delta^{(3)}(\vec{r})$.

This implies that

$$\varphi(\vec{r})\Big|^{\text{SCATT}}_{r\to\infty} = e^{ikz} - \frac{e^{ikr}}{4\pi r}\int d^3\vec{r}\,'\, u(\vec{r}\,')\, e^{i\vec{r}\cdot(\vec{k}^{(\text{in})} - \vec{k}^{(\text{out})})}$$

$$= e^{ikz} + \frac{e^{ikr}}{r} f_k(\theta,\varphi)$$

where $$f_k(\theta,\varphi) = -\frac{1}{4\pi}\int d^3\vec{r}\,'\, u(\vec{r}\,')\, e^{i\vec{r}\,'\cdot(\vec{k}^{(\text{in})} - \vec{k}^{(\text{out})})}$$

$$= -\frac{\mu}{2\pi\hbar^2}\int d^3\vec{r}\,'\, V(\vec{r}\,')\, e^{i\vec{r}\,'\cdot(\vec{k}^{(\text{in})} - \vec{k}^{(\text{out})})}.$$

Differential Cross-Section:

Observe the flux of probability out

$$j_r = \vec{j} \cdot \hat{r} = \frac{\hbar}{\mu} |\psi|^2 \frac{\partial}{\partial r} (\arg \psi)$$

with $\psi \sim \frac{e^{ikr}}{r} f_k(\theta, \varphi)$ as outgoing wave. Then

$$|\psi|^2 \sim \frac{|f_k(\theta,\varphi)|^2}{r^2}$$

$$\frac{\partial}{\partial r}(\arg \psi) \sim \frac{\partial}{\partial r}\left(kr + \arg f_k(\theta,\varphi)\right)$$

$$= k$$

So $j_r = \frac{\hbar k}{\mu} \frac{|f_k(\theta,\varphi)|^2}{r^2}$.

$ds = r^2 d\Omega$

Then $\int j_r \cdot dA_r = \int \frac{\hbar k}{\mu} \frac{|f_k(\theta,\varphi)|^2}{r^2} d\Omega = \int \frac{\hbar k}{\mu} |f_k(\theta,\varphi)| \, ds$

Now $d\sigma = \frac{\text{Flux thru } d\Omega}{\text{incoming particles}} = I \cdot dN = \text{incoming flux} \cdot \left(\text{\# particles thru } d\Omega\right)$

$$= \frac{\hbar k}{\mu}$$

Therefore $\frac{\hbar k}{\mu} \cdot \frac{\mu}{\hbar k} |f_k(\theta,\varphi)|^2 \, d\Omega = d\sigma$

$$\therefore \quad \frac{d\sigma}{d\Omega} = |f_k(\theta,\varphi)|^2 .$$